

Subhranil Deb

IMJ-PRG, Sorbonne Université, Jussieu

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EDUCATION

Institut Mathématiques de Jussieu - Paris Rive Gauche

October 2026 - present

PhD in Mathematics, Research Group : Automorphic Forms.

Advisors: Alexis Bouthier, Éric Vasserot

Université Paris Saclay

September 2025 - July 2026

M2 Analysis, Arithmetic, Geometry(AAG).

Advisor: Olivier Schiffmann

Chennai Mathematical Institute

August 2024 - May 2025

M1 Mathematics.

Chennai Mathematical Institute

August 2021 - June 2024

Bachelor of Science (Honours) in Mathematics and Computer Science.

RESEARCH INTERESTS

I am working on affine flag varieties of Kac-Moody groups and double Kazhdan-Lusztig theory.

I am in general interested in the geometric Langlands program, geometric representation theory and algebraic geometry.

PROJECTS/PRESENTATIONS

• M2 Thesis

October 2025 - Present

Instructor: Olivier Schiffmann - Université Paris Saclay

- I am currently reading about cohomological integrality for symmetric quotient stacks and the cohomological Mackey formula following the papers by Lucien Hennecart.

• Geometric Satake Equivalence

May 2025 - July 2025

Instructor: Simon Riche - Université Clermont Auvergne

- I read the theory of perverse sheaves, equivariant derived categories leading up to the Geometric Satake Equivalence. The main reference I followed is the book “Perverse Sheaves and Applications to Representation Theory” by Achar.

• Degree Bounds For Syzygies of Linearly Reductive Groups

May 2024

Instructor: Manoj Kummini - Chennai Mathematical Institute

- As a part of the course **Commutative Algebra I**, I presented “Degree Bounds For Syzygies Of Invariants” by H. Dersken. Here is the [report](#).

• Higher Dimensional Class Field Theory

May 2024

Instructor: Sukhendu Mehrotra - Chennai Mathematical Institute

- As a part of the course **Geometric Class Field Theory**, I presented “Non-abelian class field theory for arithmetic surfaces” by Hoffman, Walter and Wiesend. Here is the [report](#).

• Construction of Hilbert and Quot Schemes

May 2024

Instructor: Mohit Upmanyu and Bivas Khan - Chennai Mathematical Institute

- As a part of the course **Algebraic Geometry II**, I presented the construction of Hilbert and Quot Schemes using Castelnuovo-Mumford regularity. Here is the [report](#).

• Iwasawa Theory of Taelman Class Modules

Dec 2023

Instructor: Anwesh Ray - Chennai Mathematical Institute

- I read the book Drinfeld Modules by Mihran Papikian for my course on Function Field Arithmetic. As a part of the course **Function Field Arithmetic**, I presented Higgins thesis: “Iwasawa theory of Taelman class modules”. Here is the [report](#).

- I read R.R. Simha's paper "**The Riemann–Roch Theorem for Compact Riemann Surfaces**" for a sheaf theoretic proof of Serre Duality.
- As a part of my final presentation, I read the first 3 chapters of Diamond-Shurman's text on modular forms and presented the dimension formula as an application of the Riemann–Roch Formula.

TALKS

Perverse Sheaves and The Geometric Satake Equivalence, CMI Student Seminars([website](#)), August 2025
BGG Resolution and Higher Ext Groups in Category $\mathcal{O}$, Lie Algebra Representations Course, June 2025
Quasicoherent Sheaves on Toric Varieties, Toric Varieties Course, April 2025
Grothendieck local duality, Commutative Algebra II Course, April 2025
Clifford Algebras, Pin and Spin Groups, Lie Groups and Lie Algebra Course, Nov 2024
Galois Representations and Iwasawa Theory, CMI Student Seminars([website](#)), Jan 2024
Modular forms and Riemann-Roch Theorem, VSRP presentation, July 2023

ACHIEVEMENTS

2026 **3-year doctoral contract awarded by the École Doctorale de Sciences Mathématiques de Paris Centre (ED386) for research at IMJ-PRG,**
Sorbonne Université
2025 **Awarded the Sophie Germain Scholarship for admission in the M2 AAG program by FMJH,**
Université Paris Saclay
2024 **Selected for a fully funded research internship by [ReLaX](#) under Prof. Simon RICHE,**
Université Clermont Auvergne
2024 **Awarded Scholarship for admission in the MSc Mathematics program at CMI,**
Chennai Mathematical Institute
2023 **Selected for TIFR VSRP (Visiting Students' Research Program),** Tata Institute of Fundamental
Research (TIFR), Mumbai
2022 **Selected for Madhava Nurture camp,** S.P. College & HBCSE
2019 **Qualified Regional Mathematics Olympiad(RMO),** HBCSE
2019 **Qualified MTRP,** Indian Statistical Institute, Kolkata

CONFERENCES/WORKSHOPS

• **Derived Algebraic Geometry, CoHAs and Operads** June 2026
Université Angers
• **Enumerative geometry, rationality, derived categories and interactions** June 2026
Université Paris Saclay
• **Mathematics Inspired by Physics** June 2026
IHES
• **CARE - Collaborations in Algebra, Representation Theory and Ethics** October 2025
ENS Lyon
• **Geometry and Representation Theory around the Langlands Programs** July 2025
Université Clermont Auvergne
• **Introduction to the p-adic local Langlands program** September 2024
Indian Institute of Technology Jammu
• **Derived categories and semi-orthogonal decompositions** June 2024
Chennai Mathematical Institute
• **Rational Points on Modular Curves** September 2023
International Centre of Theoretical Sciences(Online)

TEACHING EXPERIENCE

I have been a teaching assistant for the following courses at CMI:

Topology (Instructor: Prof. Aditya Karnataki)

Algebra I (Instructor: Prof. Aditya Karnataki)

Topology (Instructor: Prof. Parameswaran Sankaran)

Analysis 3 (Instructor: Prof. Srinivasan Vasanth)

Analysis 2 (Instructor: Prof. Upendra Kulkarni)

Advanced Programming in Python (Instructor: Prof. Samir Datta)

Algebra 1 (Instructor: Prof. Upendra Kulkarni)

Introduction To Programming In Haskell (Instructor: Prof. S P Suresh)